

Chapter 25: Obstetrical Analgesia and Anesthesia — Revision Notes

Williams Obstetrics 26e, pp. 1196–1243 of the PDF. High-yield numbers are in **bold**. The eight tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: Labor starts without warning, often on a full stomach, so **aspiration** is a constant threat. Obstetrical anesthesia is now very safe: anesthesia-related maternal mortality fell **>60%** over 40 years, mainly because of **more regional analgesia**. **Neuraxial is preferred** unless contraindicated; general anesthesia carries the risk of **failed intubation**.

1. Safety and Mortality

- **2007–2017: only 0.4% of 6765 maternal deaths** were due to anesthesia complications.
- **General anesthesia deaths: ~2%** from **induction problems or failed intubation** during cesarean.
- **Regional analgesia deaths: high block (26%), respiratory failure (19%), drug reaction (19%).**
- GA's case-fatality has improved even though it is now used for the **highest-risk** women and emergencies with **decision-incision <15 min**. Fewer aspiration/hypoxic events: difficult-airway algorithms, better equipment, in-house staff.
- But **complications of regional techniques are now rising**.

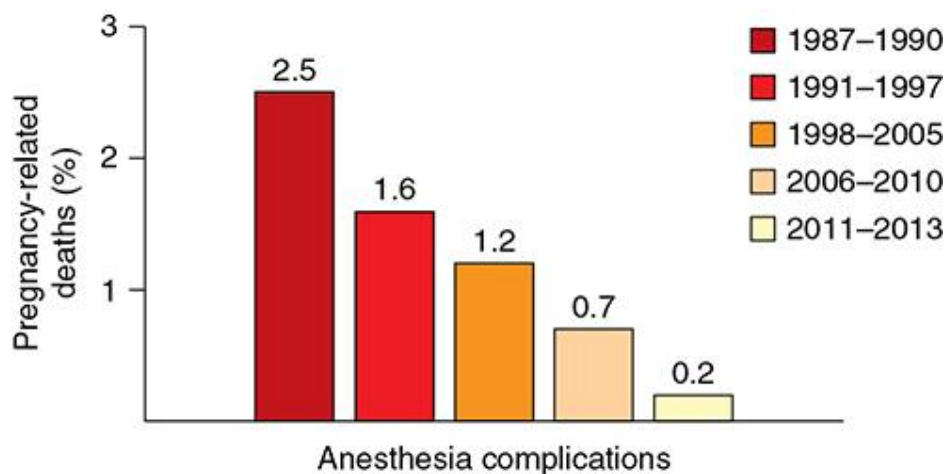


Figure 25-1. Anesthesia complications as a share of US pregnancy-related deaths: 2.5% (1987–90) → 0.2% (2011–13).

2. General Principles

Obstetrical anesthesia services

- Every obstetrician should be able to give **local and pudendal** blocks, but ideally an anesthesiologist/anesthetist provides pain relief.
- ⚠️ **A woman's request for labor pain relief is sufficient indication** (ACOG, ASA). Payers denying epidural reimbursement without a medical indication is **repudiated** by ACOG.

Table 25-1. Maternal Factors That May Prompt Anesthetic Consultation

Factor
Anatomical abnormalities of the face, neck, or spine
Thyromegaly
Cardiac, pulmonary, renal, hepatic, or neurological disease
Bleeding disorders
Prior anesthetic complications
Class III obesity (BMI ≥ 40 kg/m ²)
Obstetrical complications likely to lead to operative delivery
Severe pregnancy-associated hypertension
Obstetrical hemorrhage

BMI = body mass index.

ACOG/ASA requisites (partial list)

1. A credentialed practitioner available whenever needed, who can support vital functions in an emergency
2. Anesthesia personnel able to start a cesarean **within 30 min of the decision** (decision–incision time)
3. Anesthesia **immediately available during active labor of a TOLAC** (Ch 31)
4. A qualified anesthesiologist responsible for all anesthetics
5. A physician with obstetrical privileges for operative vaginal or cesarean delivery available during anesthesia
6. Equipment, facilities and support staff equal to a surgical suite
7. Someone other than the surgical team **immediately available to resuscitate the newborn** (Ch 32)
 - Usually needs **24-hour in-house anesthesia**; hard for small units (~40% of obstetric hospitals have <500 deliveries/yr).

Principles of pain relief

Labor phase	Pain source	Pathway
1st stage	Uterine contractions, cervical dilation	Visceral afferent sympathetic nerves, T10–L1
2nd stage	Perineal stretching	Pudendal nerve, S2–S4

- Pain perception is shaped by emotional, cognitive, social and cultural factors, expectations, age, education and support. Main causes of dissatisfaction: **perceived pain and delay in relief**.
- **Physiological effects of pain:**
 1. $\uparrow$ metabolic rate and O₂ use $\rightarrow$ **hyperventilation** $\rightarrow$ **hypocarbica**
 2. Sympathetic response $\rightarrow$ $\uparrow$ **catecholamines** $\rightarrow$ $\uparrow$ CO, SVR, BP; can impair **uterine activity and uteroplacental flow**
 3. $\uparrow$ **cortisol, endorphins**
- Effective analgesia blunts these responses.

3. Analgesia and Sedation During Labor

- Goal: discomfort only at the **acme** of contractions; rest in between. Classic combination: **meperidine + promethazine** (antiemetic tranquilizer).

Table 25-2. Typical Parenteral Analgesic Agents for Labor Pain

Agent	Usual Dose	Frequency	Onset	Neonatal Half-Life
Meperidine	25–50 mg (IV)	Every 1–2 h	5 min	~18–20 h
	50–100 mg (IM)	Every 2–4 h	30–45 min	~60 h
Fentanyl	50–100 µg (IV)	Every 1 h	1 min	~5 h
Remifentanyl	0.15–0.5 µg/kg (PCA)	Every 2 min	<1 min	10 min
Butorphanol	1–2 mg (IV or IM)	Every 4 h	1–2 min (IV); 10–30 min (IM)	~5 h
Morphine	2–5 mg (IV); 10 mg (IM)	Every 4 h	5 min (IV); 30–40 min (IM)	~7 h

IV = intravenously; IM = intramuscularly; PCA = patient-controlled analgesia. The text gives the neonatal half-life of the metabolite normeperidine as ~72 h.

Parenteral agents

Meperidine + promethazine

- **Meperidine 50–100 mg + promethazine 25 mg IM every 2–4 h**; peak analgesia **30–45 min**. IV **25–50 mg every 1–2 h** acts almost immediately.
- Most common labor opioid worldwide; better than placebo in the 1st stage.
- **Parkland IV PCA**: bolus **50 mg meperidine + 25 mg promethazine**, then **15 mg every 10 min** as needed. **Naloxone needed in 3%** of newborns.
- Crosses the placenta; fetal depression follows peak maternal analgesia. **Normeperidine neonatal half-life ~72 h** → **other opioids are now favored** (ACOG).

Butorphanol

- Opioid **agonist-antagonist**; **1–2 mg IV every 4 h**. Comparable to meperidine; **less neonatal respiratory depression**.
- Side effects: somnolence, dizziness, dysphoria.
- ⚠️ **Don't give it right after meperidine** — it antagonizes meperidine.
- Can cause a **transient sinusoidal FHR pattern** (no adverse sequelae).

Fentanyl

- **50–100 µg IV every hour**; potent but **short-acting** → frequent doses or PCA. Butorphanol gave better initial analgesia with fewer requests for more drug or epidural.

Remifentanyl

- **Ultrashort-acting**, by PCA through a **dedicated IV line** (a bolus can cause **apnea**).
- **Apnea in 25%** → respiratory monitoring and **1:1 nursing**. **Moderate hypoxia (SaO₂ <94%) in ¼** (RemiPCA SAFE).
- **Halved later epidural requests** vs IM meperidine.

Safety

- 4 of 129 anesthetic deaths were from sedation (1 aspiration, 2 inadequate ventilation, 1 overdose).
- **Naloxone** reverses neonatal opioid respiratory depression by displacing the opioid from CNS receptors. Give **after ventilation is established**.
- ⚠️ **Contraindicated in the newborn of an opioid-addicted mother** (precipitates withdrawal).

Nitrous oxide

- **Self-administered 50% N₂O in O₂** (Entonox premixed; Nitronox blender). Valve opens only on inspiration.
- **Doesn't change epidural rates** but **improves birth satisfaction**.
- Side effects: nausea, vomiting, dizziness, drowsiness. **No difference in Apgar or cord gases**.

4. Nerve Blocks — Local Anesthetic Agents

- Onset, duration and quality ↑ with dose, concentration, volume and delivery mode. **Give incremental boluses** and watch for early toxicity; resuscitation equipment must be at hand.

Table 25-3. Commonly Used Local Anesthetic Agents in Obstetrics

Anesthetic Agent ^a	Usual Concentration (%)	Usual Volume (mL)	Onset	Average Duration (min)	Maximum Dose (mg)	Clinical Use
Aminoesters ^b						
2-Chloroprocaine	2	10–20	Rapid	30–60	800	Local infiltration or pudendal block
	3	10–20		30–60	Epidural for cesarean or forceps delivery	
Aminoamides ^b						
Bupivacaine	0.0625–0.125	5–10	Slow	60–90	175	Epidural for labor
	0.5	10–20		90–150	Epidural for cesarean	
	0.75	1.5–2		60–120	Spinal for cesarean	
Lidocaine	1–1.5	10–20	Rapid	30–60	300	Local infiltration or pudendal block
	1.5–2	5–20		60–90	Epidural for cesarean	
	5	1.5–2		45–60	Spinal for D & C or puerperal tubal	
Ropivacaine	0.08–0.2	5–10	Slow	60–90	200	Epidural for labor
	0.5–1	10–30		90–150	250	Epidural for cesarean

^aWithout epinephrine. ^bEsters are hydrolyzed by plasma cholinesterases and amides by hepatic clearance. D & C = dilation and curettage. From Chestnut, 2020; Lin, 2017.

- Several concentrations and ampule sizes → **dosing errors**. Serious toxicity usually follows **inadvertent IV, subarachnoid or subdural injection**.
- Test dose (local anesthetic + epinephrine)** before an epidural:
 - Sudden ↑ HR or BP within 1–2 min → IV placement**
 - Wider-than-expected block and/or dense motor block → subarachnoid or subdural placement**

Local anesthetic systemic toxicity (LAST)

	CNS toxicity	Cardiovascular toxicity
Timing	Earlier , at lower levels	Later , at higher levels — except bupivacaine (neuro- and cardiotoxic at virtually identical levels)
Pattern	Stimulation → depression	Stimulation → depression
Features	Light-headedness, dizziness, tinnitus, metallic taste, perioral/tongue numbness , bizarre behavior, slurred speech, fasciculations → convulsions, unconsciousness	Hypertension, tachycardia → hypotension, arrhythmias , ↓ uteroplacental perfusion

Treatment

- 20% lipid emulsion (Intralipid)** pulls the drug off neural and myocardial membranes.
 - >70 kg: 100-mL bolus**, repeat **once or twice every 5 min** for persistent instability; then **200 mL over 15 min**.
- Convulsions:** airway + O₂; **lorazepam 4 mg IV slowly** (repeat once after 10–15 min) **or diazepam 5–10 mg IV every 10–15 min (max 30 mg)**. **Succinylcholine** stops the peripheral manifestations and allows intubation. **MgSO₄** (eclampsia regimen) also works.

- **Fetus:** late decelerations or bradycardia from maternal hypoxia/acidosis. **The fetus usually recovers faster in utero than after immediate cesarean** → delay delivery until maternal hypoxia and acidosis improve.
- **Hypotension:** lateral tilt, rapid crystalloid, vasopressors.
- **Cardiac arrest:** consider **bedside cesarean if no return of vital signs within 4 min**, aiming for **delivery within 5 min** (Ch 50).
- Bupivacaine is mostly used dilute epidurally; **0.75% is limited to small spinal doses.**

5. Pudendal Block

- **Pudendal nerve (S2–S4 ventral branches)** supplies the perineum, anus, vulva and clitoris. It passes **beneath the sacrospinous ligament where it attaches to the ischial spine.**
- **Technique:**
 1. Tubular introducer guides a **15-cm, 22-gauge needle**; introducer tip against the vaginal mucosa **just below the ischial spine**, needle protruding **1.0–1.5 cm**
 2. Mucosal wheal with **1 mL of 1% lidocaine**
 3. **Aspirate before every injection**
 4. Advance to the **sacrospinous ligament**, infiltrate **3 mL lidocaine**
 5. Repeat on the other side
- **Effective within 3–4 min** (pinch of lower vagina and posterior vulva painless).
- If delivery comes first and an episiotomy is needed: infiltrate **5–10 mL of 1% lidocaine** at the site.
- **Lidocaine maximum 4.5 mg/kg, not >300 mg.** Example: $50 \text{ kg} \times 4.5 = 225 \text{ mg} = 22.5 \text{ mL of 1\%}$. **1% = 10 mg/mL.**
- **Not adequate** for extensive manipulation, visualizing the cervix/upper vagina, or **manual uterine exploration** → parenteral or neuraxial needed.
- **Complications:** IV injection (toxicity); **hematoma** (esp. coagulopathy); rare severe infection spreading to the **hip joint, gluteal muscles or retrosoas space**. Adding corticosteroid did **not** reduce pudendal neuralgia at 3 months.

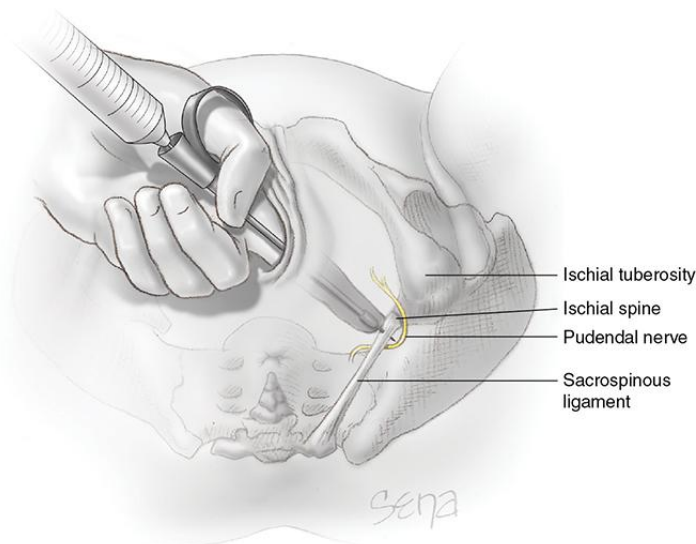


Figure 25-2. Transvaginal pudendal block: needle beyond the guard, through the sacrospinous ligament just below the ischial spine.

6. Paracervical Block

- **Uterovaginal plexus** lies lateral to the uterosacral ligaments → inject just lateral to their uterine insertion.
- **1% lidocaine or 3% chloroprocaine, 5–10 mL at 4 and 8 o'clock.**

- Relieves **1st-stage** pain only (pudendal not blocked); short-acting, may need repeating.
- ⚠️ **Fetal bradycardia in ~15%**, within **10 min**, lasting **up to 30 min** — likely **uterine artery vasospasm** (↑ pulsatility index).
- **Avoid with possible fetal compromise. Not used at Parkland.**

7. Neuraxial Analgesia

Overview

- Epidural, spinal, dural-puncture epidural (DPE), combined spinal–epidural (CSE).
- **~75%** of US mothers receive neuraxial analgesia. First births **68%** vs later births **57%** (vaginal deliveries).
- Antepartum education ↑ use by **30%** in Hispanic women.
- Use rises with obesity class: **I 72%, II 73%, III 76%**.
- **Anatomy:** epidural space lies just outside the dura; the **subarachnoid space** (inside the dura) holds the cauda equina and CSF. **Spinal = intrathecal.**
- **Epidural** = catheter, mainly for labor (also operative delivery). **Spinal** = single intrathecal injection, usually for operative vaginal or cesarean delivery.

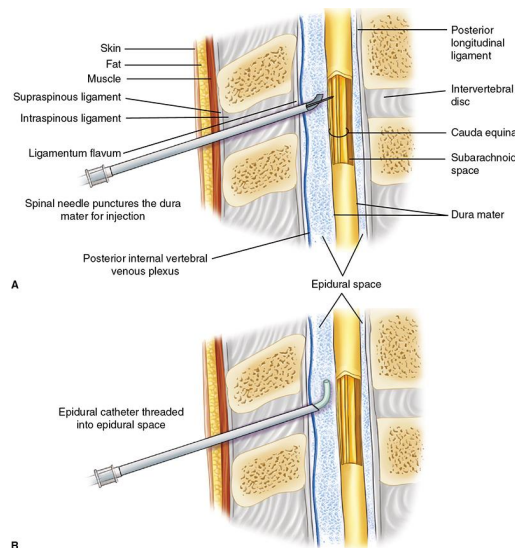


Figure 25-3. (A) Combined spinal–epidural: spinal needle through the epidural needle punctures the dura. (B) Epidural catheter threaded into the epidural space.

8. Spinal (Subarachnoid) Block

- Advantages: **short procedure, rapid onset, high success.**
- **Subarachnoid space is smaller in pregnancy** (engorged internal vertebral venous plexus) → the **same dose gives a higher block** than when nonpregnant.
- **Hyperbaric** solution (with dextrose) is denser than CSF: **sitting** → **settles caudally**; lateral → more effect on the dependent side.

Use	Level	Agent/dose
Operative vaginal delivery	T10 (umbilicus)	Hyperbaric lidocaine (rapid, short) or isobaric bupivacaine (>1 h). Given at full dilation once forceps criteria are met. Pre-hydrate with 1 L crystalloid
Cesarean	T4	Hyperbaric bupivacaine 10–12 mg + fentanyl 20–25 µg (better, longer block; less shivering and referred pain) ± intrathecal morphine 0.1–0.2 mg (postoperative pain)

Intrathecal morphine monitoring (SOAP)

- **Low dose (>50 to ≤150 µg), no risk factors:** respiratory rate and sedation checks for **12 h**.
- **Additional risk factors:** checks for **24 h** ± pulse oximetry or capnography.
- **Dose >150 µg:** same as with risk factors (24 h ± oximetry/capnography).

Table 25-4. Complications of Neuraxial Analgesia

Complication
Fever >38°C
Hypotension — highest with spinal analgesia (~50%)
Fetal heart rate abnormalities
High spinal level
Increased QT interval — with spinal analgesia in postterm pregnancies
Inadequate analgesia (~10%)
Postdural puncture headache (1–3%)
Nerve injury — arachnoiditis
Pruritus — with addition of opioids

From ACOG, 2019b; ASA, 2016; D'Angelo, 2014; Karahan, 2020; Maronge, 2018. The table's PDPH rate (1–3%) is higher than the text's figure for spinal analgesia (0.2–1%).

Hypotension

- **Sympathetic blockade** → **vasodilation**, worsened by **caval compression**. Placental flow may fall even without maternal hypotension when supine.
- **Treatment: left lateral uterine displacement**, rapid IV crystalloid, **IV vasopressors** (a systematic review: give vasopressors only if hypotensive).
- **Phenylephrine = preferred** (pure α-agonist):
 - **Better cord gases / less fetal acidosis** than ephedrine (meta-analysis of 36 RCTs: equal efficacy)
 - **More maternal bradycardia** than ephedrine or norepinephrine
 - **Prophylactic infusion** works and needs less intervention than boluses
- **Norepinephrine** may ↑ CO (β effect) but gives a **lower cord arterial pH** than phenylephrine.

High (total) spinal blockade

- Usually from an **excessive dose**; also from a small volume in the **subdural space** (between dura and arachnoid; small, so it spreads high).
- **Rapid hypotension and apnea** → treat at once to prevent arrest:
 1. **Lateral uterine displacement**
 2. **Ventilation, preferably intubation**
 3. **IV fluids + vasopressors**

Postdural puncture headache (PDPH)

- **CSF leak** → traction on pain-sensitive structures when upright; or compensatory cerebral vasodilation (**Monro-Kellie**).
- **Spinal: 0.2–1%** (overall ~1 per 200 blocks). **40%** of postpartum headache visits are from neuraxial anesthesia, mostly **within 7 days**.
- **Prevention: small-gauge, pencil-point needles (Sprotte, Whitacre)**, avoid multiple punctures.
- **⚠ No benefit** from lying flat, vigorous hydration, or a **prophylactic blood patch** within 24 h.
- **Caffeine** (cerebral vasoconstrictor): limited efficacy.

- **Severe headache** → **epidural blood patch (most effective): 10–20 mL autologous blood** injected **at or below** the puncture site (spreads cephalad). Immediate relief; few complications.
- **Non-postural or persistent after a repeat patch** → **think of another cause: superior sagittal sinus thrombosis**, persistent CSF leak, **pneumocephalus**, intracranial or intraspinal **subarachnoid hematoma**.

Other complications

- **Convulsions with PDPH:** rare, with temporary blindness (8 in 19,000 at Parkland); presumed from CSF-leak hypotension. Treat seizures + blood patch.
- **Bladder dysfunction:** obtunded sensation → postpartum **distention** (especially with large IV volumes). **Intermittent catheterization** → **more bacteriuria than continuous** (Millet, 146 women).
- **Arachnoiditis and meningitis:** rare thanks to **preservative-free (no methylparaben)** agents, asepsis and disposable kits.

Table 25-5. Absolute Contraindications to Spinal Analgesia

Contraindication
Refractory hypotension
Maternal coagulopathy
Thrombocytopenia <70,000/ μ L
Untreated maternal bacteremia
Skin infection at site of needle placement
Increased intracranial pressure
Prophylactic low-molecular-weight heparin within 12 hours

Other considerations

- **Hypovolemia/hemorrhage** → spinal contraindicated (additive cardiovascular effects).
- **Severe preeclampsia:** BP may fall → **judicious dosing**. Preeclamptic women under GA needed more antihypertensives than with spinal.
- **Bleeding diatheses (von Willebrand, hemophilia):** ↑ subarachnoid hematoma risk, but neuraxial is **safe after factor replacement and normal levels**. Consult anesthesia before labor.
- **Anticoagulation (SOAP):** UFH >10,000 U/d → individual assessment; **prophylactic LMWH** → **12-h delay**; **therapeutic LMWH** → **24-h delay** (Ch 55).
- **Relative contraindications:** **Chiari malformation, pseudotumor cerebri** (often OK after anesthesia + neurology consultation); **severe aortic stenosis, pulmonary hypertension** (can't tolerate ↓ preload/afterload).

9. Epidural Analgesia

- Epidural space: areolar tissue, fat, lymphatics, **internal vertebral venous plexus**. Usually **lumbar** entry with an indwelling catheter → boluses or continuous infusion (patient- or caregiver-controlled). Trained L&D nurses may adjust or stop infusions under physician supervision.

Indication	Block needed
Labor + vaginal delivery	T10–S4
Cesarean	T4–S1

- Spread depends on catheter tip, dose, concentration and volume. **Septa/synechiae** or catheter **migration** can give a patchy block.
- Maintain with continuous infusion or **programmed intermittent boluses**. Adding **fentanyl or sufentanil** reduces local anesthetic → **preserves motor function**. Popular regimen: **bupivacaine 0.0625–0.125% + fentanyl 2 μ g/mL**.

Table 25-6. Technique for Labor Epidural Analgesia

Step
Obtain informed consent and consult the obstetrician
Monitoring: BP every 1–2 min for 15 min after a local anesthetic bolus; continuous maternal HR during induction; continuous FHR; continual verbal communication
Hydration with 500–1000 mL of Ringer lactate
Woman in a lateral decubitus or sitting position
Epidural space identified by loss of resistance
Catheter threaded 3–5 cm into the epidural space
After negative aspiration, test dose: 3 mL of 1.5% lidocaine with 1:200,000 epinephrine , injected between contractions (after careful aspiration and after a contraction, so labor-pain tachycardia is not confused with IV-injection tachycardia)
Positive test dose (IV): tinnitus, circumoral numbness, dizziness, HR ↑ >20 bpm within 1 min. Subarachnoid: inability to raise the legs against gravity within 4 min
If negative: 10–15 mL of 0.0625–0.125% bupivacaine to reach a T10 sensory level
After 15–20 min , assess block (cold or pinprick). No block → replace catheter. Asymmetric → withdraw 0.5–1.0 cm and give 3–5 mL of 0.25% bupivacaine ; still inadequate → replace
Position lateral or semilateral (avoid aortocaval compression)
Then BP every 5–15 min ; continuous FHR
Level of analgesia and motor block assessed at least hourly

Complications of epidural analgesia

- **High spinal block:** from unrecognized subarachnoid or subdural injection. **Accidental dural puncture 0.91%** (>18,000 women).
- **Ineffective analgesia:**
 - **90%** rate relief good to excellent; **12%** have ≥3 episodes of pain/pressure; efficacy ↓ with rising BMI.
 - Consider **replacing the catheter**. A poorly working catheter can't be extended for cesarean: **conversion fails ~20%** → new block or GA. MFMU: **4%** of women with labor epidurals needed **GA for cesarean**.
 - Poor perineal analgesia → extra doses or **add a pudendal block**.
- **Hypotension:**
 - **Most frequent side effect; needs treatment in 1/3.**
 - **20% if admission pulse pressure <45 mm Hg vs 6% if >45;** ↑ in obesity.
 - Prevent with **500–1000 mL crystalloid** and lateral position.
- **CNS stimulation/convulsions:** rare → lipid emulsion + lorazepam or diazepam.
- **Fever:**
 - **10–15% above baseline** rate.
 - Theories: **infection vs thermoregulatory dysregulation**.
 - Fever only with **placental inflammation** (Dashe) → suggests infection; but **cefoxitin 2 g prophylaxis did not reduce fever (~40% in both arms;** Sharma, 400 nulliparas) → infection unlikely.
 - Other ideas: hypothalamic set-point change, blocked warm-receptor input, heat imbalance, **bupivacaine blocking antipyretic interleukins**.
 - Persistent fever → usually treated as **presumed chorioamnionitis**.
- **Back pain:** common short-term; **no link to new long-term backache**.
- **Miscellaneous:**
 - **Spinal/epidural hematoma** (rare, neurological emergency → usually surgery).
 - **Epidural abscess (Staphylococcus aureus;** IV antibiotics ± surgery) → early neurosurgery consult.

- **Sheared catheter:** locate by **CT**; asymptomatic tip in the epidural space → surveillance; neurological signs or subarachnoid fragment → neurosurgery.
- **Autism link unlikely.**

Effect on labor

- Epidural **prolongs labor and ↑ oxytocin use** (Parkland pooled RCTs).
- Alexander (459 nulliparas): active phase **~1 h longer** than Friedman.
- Systematic review (Anim-Somuah 2018): **1st stage +32 min, 2nd stage +15 min.**
- 2nd-stage prolongation was worse with the older, denser blocks → more operative vaginal delivery. Craig (310): **second-stage bupivacaine caused motor block but did not lengthen the 2nd stage**; outcomes and satisfaction the same as fentanyl alone.
- Spinal-epidural and epidural give similar labor durations.

Table 25-7. Selected Labor Events in 2703 Nulliparous Women Randomized to Epidural Analgesia or Intravenous Meperidine Analgesia

Event ^a	Epidural Analgesia (n = 1339)	Intravenous Meperidine (n = 1364)	p value
Labor outcomes			
First-stage duration (h) ^b	8.1 ± 5	7.5 ± 5	0.011
Second-stage duration (min)	60 ± 56	47 ± 57	<0.001
Oxytocin after analgesia	641 (48)	546 (40)	<0.001
Type of delivery			
SVD	1027 (77)	1122 (82)	<0.001
Forceps	172 (13)	101 (7)	<0.001
Cesarean	140 (10.5)	141 (10.3)	0.92

^aData are presented as n (%) or mean ± SD. ^bFirst stage = initiation of analgesia to complete cervical dilation. SVD = spontaneous vaginal delivery.

Fetal heart rate and neonate

- Epidural (0.25% bupivacaine) vs IV meperidine: **no worrisome FHR effects; meperidine → ↓ variability and fewer accelerations.**
- Epidural → **better neonatal acid–base status** than meperidine.

Cesarean delivery rates

- Early data showing more cesareans came from the **dense motor block** era. With **dilute solutions, cesarean risk is NOT increased** (ACOG, ASA).

Table 25-8. Cesarean Delivery (CD) Rates in Laboring Women Receiving Epidural Analgesia or Intravenous Meperidine

Study	Odds Ratio ^a	CD Incidence Higher with
Ramin (1995)	1.20 (0.73–1.97)	Epidural
Sharma (1997)	0.77 (0.31–1.91)	Meperidine
Gambling (1998)	1.13 (0.65–1.97)	Epidural
Lucas (2001)	1.05 (0.68–1.63)	Epidural
Sharma (2002)	0.81 (0.41–1.61)	Meperidine
Adjusted	1.04 (0.81–1.34)	—

^aShown with 95-percent confidence intervals.

- Every confidence interval crosses 1 → **epidural does not significantly change the cesarean rate.**

Timing of placement

- Early observational data linked early epidurals to cesarean, but **≥5 RCTs: timing has no effect on cesarean, forceps or malposition.**
- **⚠ Withholding an epidural until an arbitrary dilation only denies pain relief.**

Safety

- MFMU (~20,000 epidurals): **no anesthesia-related maternal deaths.**
- Ruppen (27 studies, **1.4 million** women):

Event	Rate
Deep epidural infection	1 : 145,000
Epidural hematoma	1 : 168,000
Persistent neurological injury	1 : 240,000

- **SOAP Serious Complication Repository (>228,000 neuraxial, 5 yr):** no anesthesia deaths; **high neuraxial block 1 in 4000** (58); **respiratory arrest 1 in 10,000** (16); 27 neurological injuries; 4 abscess/meningitis.

Contraindications (see Table 25-5)

Thrombocytopenia

- The platelet count at which epidural bleeding occurs is unknown. Epidural hematoma **0–0.6%** with platelets **<70,000/μL**.
- **ACOG/SOAP: platelets ≥70,000/μL → may have neuraxial.**

Anticoagulation (ACOG 2019b)

Regimen	Neuraxial placement
Prophylactic UFH 5000 U twice daily	Not contraindicated
Intermediate UFH (7500–10,000 U)	Likely low risk >12 h after last dose
High-dose UFH (>20,000 U)	Likely low risk >24 h after last dose if aPTT normal or anti-Xa undetectable
Prophylactic LMWH	Delay placement or removal ≥12 h
Therapeutic LMWH	Delay placement or removal 24 h

Severe preeclampsia–eclampsia

- **Epidural is favored over GA.**
 - Epidural concerns: hypotension, pressor-induced hypertension, **pulmonary edema** from large crystalloid volumes.
 - GA risks are worse: **difficult intubation (airway edema)**, **sudden severe hypertension** → pulmonary/cerebral edema, **intracranial hemorrhage**.
- **Lucas (738 hypertensive women, Parkland RCT):** labor epidural is **safe** with a standard protocol (prehydration, incremental dosing, ephedrine).
- **HELLP/thrombocytopenia:** neuraxial OK if **platelets ≥70,000/μL**.
- **Contracted intravascular volume but ↑ total body water** (capillary leak) → aggressive fluids risk **pulmonary edema, especially in the first 72 h postpartum.**
 - Unlimited preload → pulmonary edema in **3.5%** (Hogg).
 - **Preload limited to 500 mL → no pulmonary edema** (Lucas). Usual judicious preload **500–1000 mL**.
- Slow onset with dilute solutions → less abrupt vasodilation.
- Vigorous crystalloid → **cerebral edema** and **pharyngolaryngeal edema**.

10. Other Neuraxial and Regional Techniques

Technique	How	Pros	Cons / key data
Epidural opioid	Opioid with a local anesthetic (opioid alone usually inadequate)	Less local anesthetic → less motor block , rapid onset, less shivering	Pruritus, urinary retention → relieved by nalbuphine without losing analgesia. Monitor respiration with neuraxial morphine
Combined spinal–epidural (CSE)	Needle-through-needle: intrathecal opioid ± local anesthetic, then epidural catheter	Rapid, profound relief with virtually no motor block	Outcomes similar to epidural (Miro, 6497 women). ↑ FHR abnormalities from uterine hypertonus ; hypotension, pruritus. Intrathecal sufentanil ↓ contractility but not labor length. Failure: CSE 6.6% vs epidural 11.6%
Continuous spinal	Intrathecal catheter	—	Seldom used (PDPH). Redesigned kits: complications = epidural (Arkoosh, 429)
Dural-puncture epidural (DPE)	25-gauge spinal needle punctures the dura (CSF drip confirms), nothing injected ; then epidural catheter	Faster onset than epidural, fewer side effects than CSE , better sacral coverage , fewer top-ups	PDPH not significantly increased (5 studies, 581 women)

Local infiltration for cesarean

- To supplement a **patchy** emergency block, or rarely for an emergency cesarean with **no anesthesia available**.
- Method 1 — layer-by-layer:** skin along the incision, then subcutaneous, muscle and rectus sheath as the abdomen is opened. **Avoid large volumes in fat** (few nerves). Cooper: **52 mL of 0.5% lidocaine with 1:200,000 epinephrine = 260 mg**. With **epinephrine**, lidocaine **max 7 mg/kg** (vasoconstriction slows absorption). 1% procaine also used.
- Method 2 — field block:**
 - 10th–12th intercostal nerves:** midway between costal margin and iliac crest in the **midaxillary line** → needle medially down to the transversalis fascia, **5 mL of 0.5% lidocaine with epinephrine**, then redirect 45° cephalad and caudad
 - Ilioinguinal and genitofemoral:** at the **external inguinal ring**, **2–3 cm lateral to the pubic tubercle** at 45°
 - Both sides (4 skin punctures total), then the **skin over the incision**

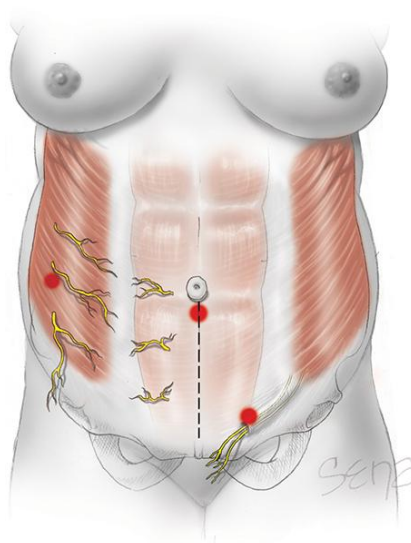


Figure 25-4. Local block for cesarean: midaxillary point (T10–T12 intercostals), external inguinal ring (ilioinguinal, genitofemoral), and the incision line.

Transversus abdominis plane (TAP) block

- **Ultrasound-guided** injection between the **internal oblique and transversus abdominis** → **T6–L1** anterior abdominal wall.
- High vs low dose similar. For postoperative pain: **not better than wound infiltration**; comparable to IV PCA; **equivalent to intrathecal morphine**. **Liposomal bupivacaine** incisional block **no better than placebo**.

11. General Anesthesia

- Needs trained staff and special equipment (incl. **fiberoptic intubation**).
- **Failed intubation ~1 in 390** obstetric GAs → **neuraxial preferred** unless contraindicated.
- **93% of >54,000 cesareans** (MFMU) used neuraxial. GA used more in **non-white women**; common for **emergent cesarean for fetal distress**. Obstetric anesthesiologists use neuraxial more.

Patient preparation

1. **Antacid** — probably lowered GA mortality **more than any other single practice**. ASA: **nonparticulate antacid, H₂-blocker or metoclopramide**. Parkland: **30 mL Bicitra (sodium citrate + citric acid)** a few minutes before GA or major neuraxial block; **repeat if >1 h** passes before induction.
2. **Lateral uterine displacement** (duration of GA then matters less for the neonate).
3. **Preoxygenation**: ↓ FRC and ↑ O₂ use → **rapid desaturation during apnea** (worse with obesity). **100% O₂ by face mask for 2–3 min**, or in an emergency **4 vital-capacity breaths**. **High-flow nasal O₂ is inferior** to face mask.

Induction

Agent	Note
Propofol	Hypotension
Etomidate	Fewer hemodynamic changes
Ketamine	For hemodynamically unstable women

Intubation

- **Succinylcholine** (rapid, short-acting) right after loss of consciousness.
- **Cricoid pressure (Sellick maneuver)** from induction until the tube is placed; confirm tube position before surgery.
- **Predictors of difficulty**: prior difficult intubation, neck/maxillofacial/pharyngeal/laryngeal anatomy, **intrapartum airway edema, morbid obesity** (major risk).
- Have ready: different laryngoscopes, **laryngeal mask airway, video laryngoscope**, transtracheal ventilation kit, **awake fiberoptic** intubation.
- **Failed intubation algorithm**:
 1. **Mask ventilation + cricoid pressure**
 2. Proceed with mask ventilation **or let her wake up**
 3. **Can't ventilate** (oral airway, LMA, fiberoptic fail) = life-threatening → **percutaneous or open cricothyrotomy + jet ventilation - Failed intubation drills** recommended.

Inhalational agents

- **Desflurane, sevoflurane** (volatile halogenated) → amnesia and analgesia.
- **High concentrations relax the uterus** → used for **internal podalic version of twin B, breech decomposition, EXIT procedures, acute uterine inversion**.

Extubation

- Only when **awake, following commands and maintaining saturation**. Consider **NG emptying** of the stomach first.
- **Now relatively more dangerous than induction**: Michigan 1985–2003 — of 15 anesthesia deaths, **none at induction, 5 from hypoventilation/obstruction at emergence, extubation or recovery**.

Fasting

- **Uncomplicated labor: modest clear liquids allowed** (water, clear tea, black coffee, carbonated drinks, pulp-free juice). **ERAS: clear liquids up to 2 h** before scheduled surgery.
- **Avoid solid foods. Elective cesarean or puerperal tubal: fast 6–8 h** depending on food type.
- Gastric fluid in nonlaboring women (**~2 mL/kg**) $\approx$ nonpregnant, but pregnancy shifts the stomach horizontally and $\downarrow$ esophageal sphincter tone.
- Parkland: clear liquids for low-risk laboring women expecting vaginal delivery.

12. Aspiration

- **Aspiration pneumonitis** — historically the **most common cause of obstetric anesthetic death**.
- **Prevention**: routine **antacids, cricoid pressure** with intubation, **regional analgesia** whenever possible.

Pathophysiology

- **Aspirate pH <2.5** $\rightarrow$ **severe chemical pneumonitis** (Teabeaut 1952). **Nearly half** of laboring women have gastric pH <2.5.
- **Right mainstem bronchus** $\rightarrow$ **right lower lobe** most often; severe cases bilateral.
- Onset immediate or hours later.

Aspirated material	Result
Large solid material	Airway obstruction
Small particles, non-acidic	Patchy atelectasis $\rightarrow$ bronchopneumonia
Highly acidic liquid	$\downarrow$ SaO ₂ , tachypnea, bronchospasm , rhonchi, rales, atelectasis, cyanosis, tachycardia, hypotension; capillary leak of protein- and RBC-rich fluid $\rightarrow$ $\downarrow$ compliance, shunt, severe hypoxemia

- **Earliest and most sensitive signs: respiratory rate and pulse oximetry**. X-ray changes may lag $\rightarrow$ **a normal chest radiograph does not exclude aspiration**.

Treatment

- **Wipe the mouth and suction** pharynx and trachea at once.
- **⚠ No saline lavage** (spreads the acid).
- **Bronchoscopy** if large particles obstruct the airway.
- **⚠ No corticosteroids or prophylactic antibiotics**. Treat infection vigorously if it develops (now community- or hospital-type organisms rather than anaerobes; Ch 54).
- **ARDS** $\rightarrow$ **mechanical ventilation with PEEP** (Ch 50).

Quick-Recall: Choosing and Troubleshooting Analgesia

Situation	Choice / action
Mild-moderate labor pain, no neuraxial	Meperidine + promethazine; butorphanol; fentanyl; remifentanyl PCA (1:1 nursing); N ₂ O
Opioid-exposed newborn with respiratory depression	Ventilate, then naloxone — not if mother is opioid-addicted
Outlet/low forceps or episiotomy repair	Pudendal block (S2–S4) or low spinal to T10
First-stage pain, no anesthesiologist	Paracervical block — but 15% fetal bradycardia; avoid if fetal compromise
Labor analgesia	Epidural T10–S4 (bupivacaine 0.0625–0.125% + fentanyl 2 µg/mL); CSE for fastest onset; DPE for better sacral coverage
Cesarean	Spinal T4: hyperbaric bupivacaine 10–12 mg + fentanyl 20–25 µg ± morphine 0.1–0.2 mg
Hypotension after block	Lateral tilt, crystalloid, phenylephrine
Test dose: HR ↑ >20 bpm in 1 min	IV catheter → resite
Can't lift legs in 4 min / wide dense block	Subarachnoid/subdural → watch for high spinal
High spinal	Displace uterus, intubate/ventilate, fluids + pressors
LAST (tinnitus, perioral numbness → seizures/arrhythmia)	20% lipid 100 mL bolus (repeat ×1–2 q5 min) → 200 mL/15 min; lorazepam 4 mg or diazepam; arrest → deliver by 5 min
Severe postural PDPH	Epidural blood patch 10–20 mL at/below the site
Platelets <70,000/µL	No neuraxial
Prophylactic LMWH / therapeutic LMWH	Wait 12 h / 24 h
Severe preeclampsia	Epidural preferred; preload ≤500–1000 mL
General anesthesia	Bicitra 30 mL, lateral tilt, preoxygenate 2–3 min, propofol/etomidate/ketamine + succinylcholine, cricoid pressure
Failed intubation, can't ventilate	Cricothyrotomy + jet ventilation
Need uterine relaxation	High-concentration volatile agent (desflurane/sevoflurane)
Aspiration	Suction; no lavage, no steroids, no prophylactic antibiotics; PEEP for ARDS

Key Numbers to Memorize

- Anesthesia = **0.4%** of maternal deaths (2007–17); regional deaths: high block **26%**, respiratory failure **19%**, drug reaction **19%**
- Decision-incision capability **30 min**; **40%** of units <500 births/yr
- Pain pathways: **T10–L1** (1st stage), **S2–S4** (2nd stage)
- Meperidine IM **50–100 mg** + promethazine **25 mg** q2–4 h; normeperidine neonatal t_{1/2} **~72 h**; naloxone needed **3%** (PCA)
- Remifentanyl: apnea **25%**, SaO₂ <94% in **¼**
- Lidocaine max **4.5 mg/kg (≤300 mg)**; **7 mg/kg with epinephrine**; **1% = 10 mg/mL**
- Lipid rescue: **100 mL bolus** (>70 kg) → **200 mL/15 min**; lorazepam **4 mg**; diazepam **5–10 mg (max 30)**
- Perimortem cesarean: start at **4 min**, deliver by **5 min**
- Pudendal: **15-cm 22-G** needle, **1 + 3 mL** per side, works in **3–4 min**
- Paracervical: **4 and 8 o'clock**, **5–10 mL**; bradycardia **15%** (within 10 min, up to 30 min)
- Neuraxial use **~75%**
- Spinal: **T10** (vaginal), **T4** (cesarean); bupivacaine **10–12 mg**, fentanyl **20–25 µg**, morphine **0.1–0.2 mg**; monitor **12 h / 24 h**

- Epidural: **T10–S4** labor, **T4–S1** cesarean; test dose **3 mL 1.5% lidocaine + 1:200,000 epi**; catheter **3–5 cm**
- Hypotension: spinal **~50%**; epidural needs treatment in **1/3**; **20% vs 6%** by pulse pressure **<45 / >45 mm Hg**
- PDPH spinal **0.2–1%** (~1/200); accidental dural puncture **0.91%**; blood patch **10–20 mL**
- Epidural: good relief **90%**; conversion failure **~20%**; GA needed **4%**; fever **+10–15%**
- Labor: 1st stage **+32 min**, 2nd **+15 min**; cesarean **10.5% vs 10.3%**; adjusted OR **1.04**
- Rare events: infection **1:145,000**, hematoma **1:168,000**, persistent injury **1:240,000**; high block **1:4000**; respiratory arrest **1:10,000**
- Platelets **≥70,000/μL**; LMWH **12 h / 24 h**; UFH 5000 bid OK; 7500–10,000 → **>12 h**; >20,000 → **>24 h**
- CSE failure **6.6%** vs epidural **11.6%**; DPE needle **25-G**
- Failed intubation **1 in 390**; neuraxial for **93%** of cesareans; preoxygenate **2–3 min** or **4 breaths**; Bicitra **30 mL**
- Aspiration: **pH <2.5**; right lower lobe; elective surgery fast **6–8 h** (clear liquids to **2 h**)